WindFlow airflow study

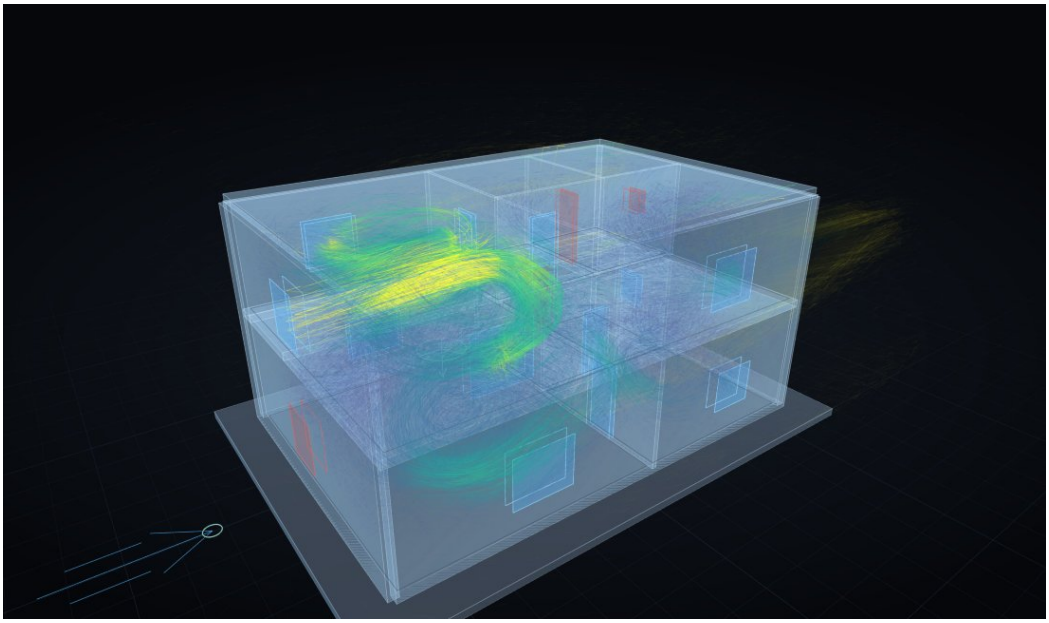
Sample house

wind 4.0 m/s from 270° (W), 12 of 16 openings open

Prepared by WindFlow headless check · 29 October 2026

Summary

With a wind of 4.0 m/s from 270° (W) and 12 of 16 openings open, the building gets 15.0 fresh-air air changes per hour. Bedroom 1 is the best ventilated room (99.0) and Bedroom 2 the least (8.9). Bathroom has no air path with the current openings. This is a Preview result: use it for direction only.



Key numbers

Result	Preview: a fast, coarse run, good for direction only
Wind	4.0 m/s from 270° (W) at 10 m height
Quality / backend	Preview · CUDA · NVIDIA GeForce RTX 5070 Ti
Grid	166 × 150 × 62 cells, 0.25 m
Steps / solve time	2,832 steps · 1.6s
Rooms analysed	7
Best ventilated	Bedroom 1 — 99.0 fresh-air ACH
Least ventilated	Bathroom — sealed (no air path)
Building fresh-air rate	15.0 ACH (3600 ÷ mean air age over the indoor volume without the sealed room Bathroom (23.4 m^3 at the 24 h age cap))
Total inflow	4.72 m^3/s (16984 m^3/h)

Mass balance OK, residual 2 %

See Notes at the end for where this result is less certain.

Plan · L1 · 1.1 m

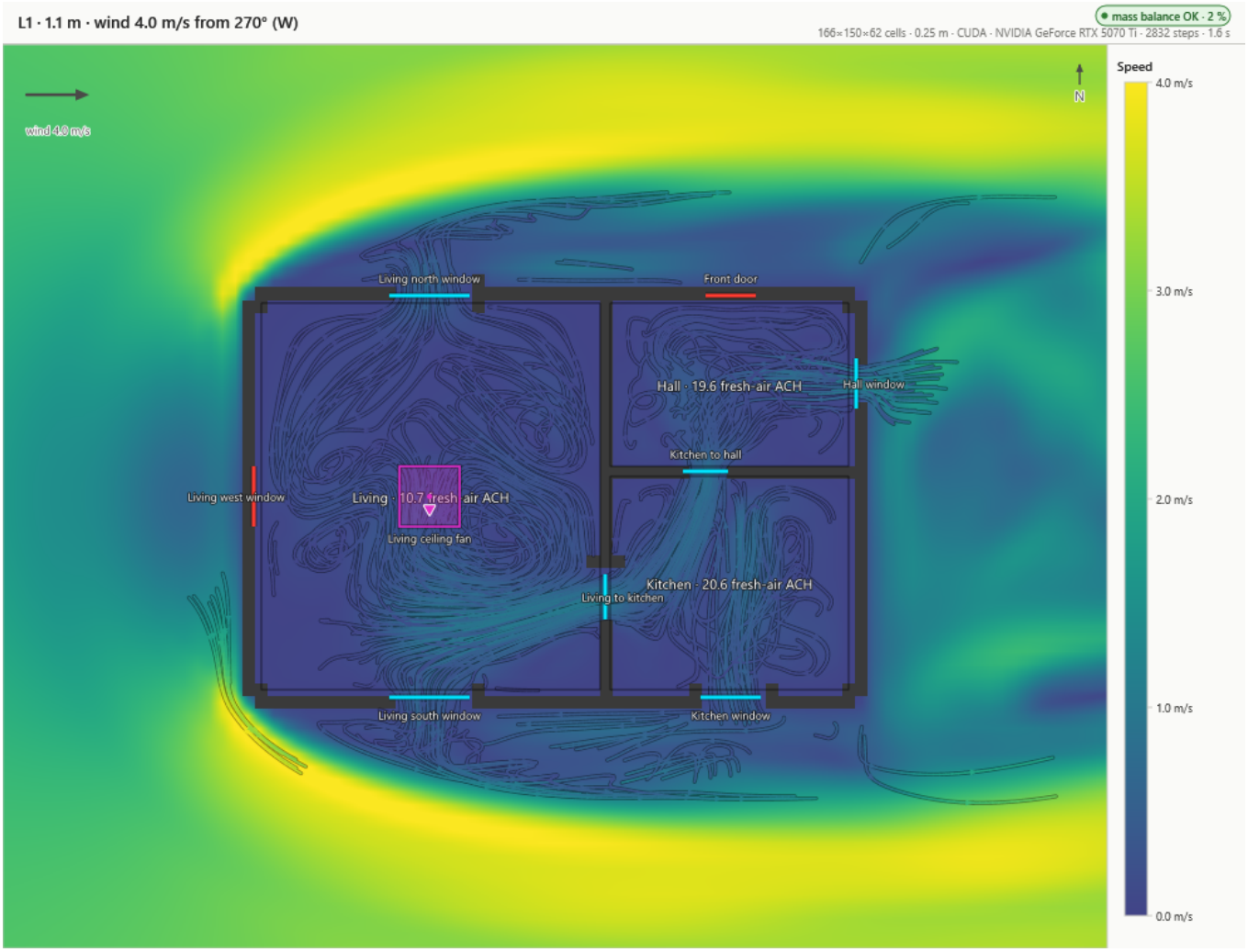


Figure 1: Plan · L1 · 1.1 m

Plan · L2 · 4.3 m

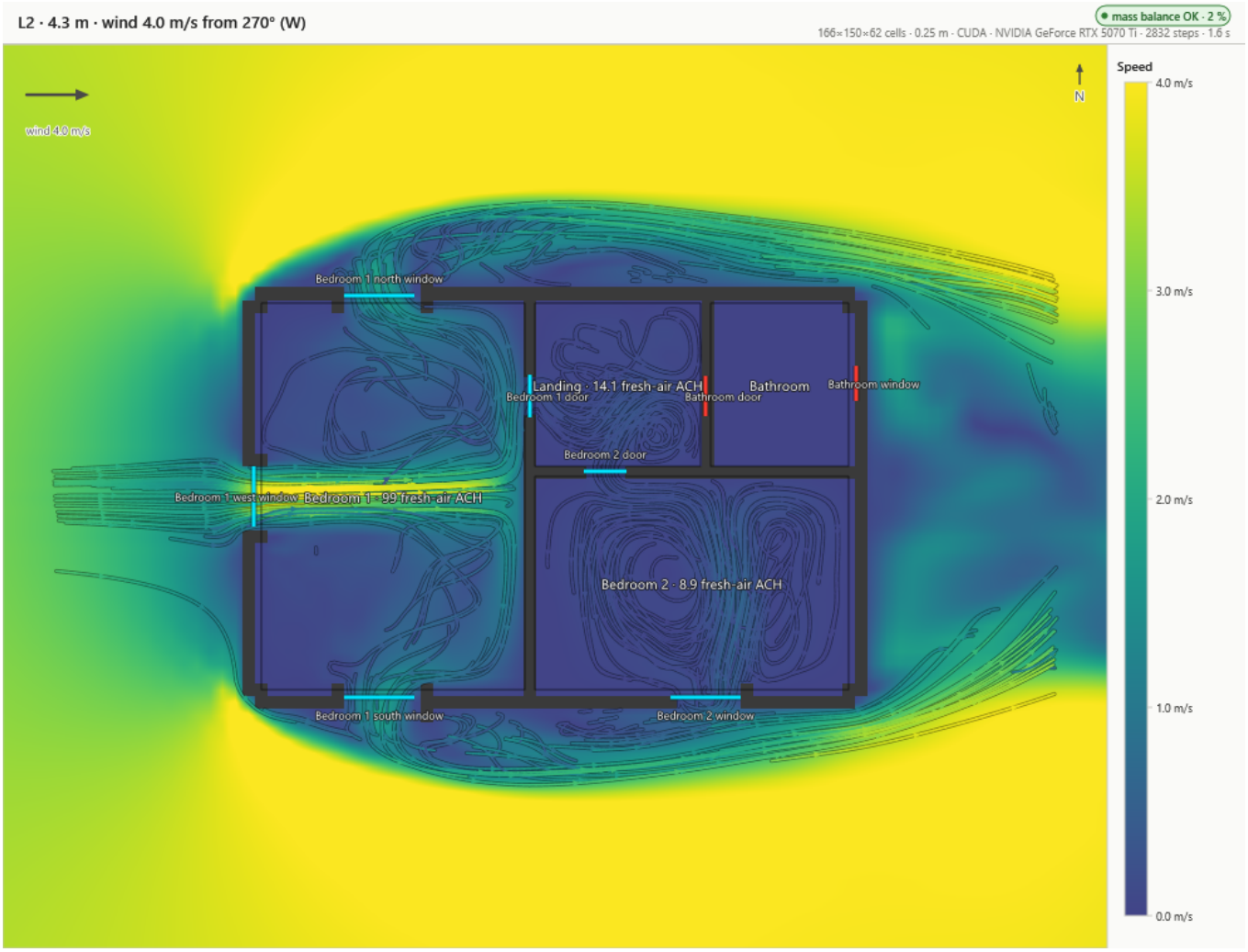


Figure 2: Plan · L2 · 4.3 m

Section · long axis

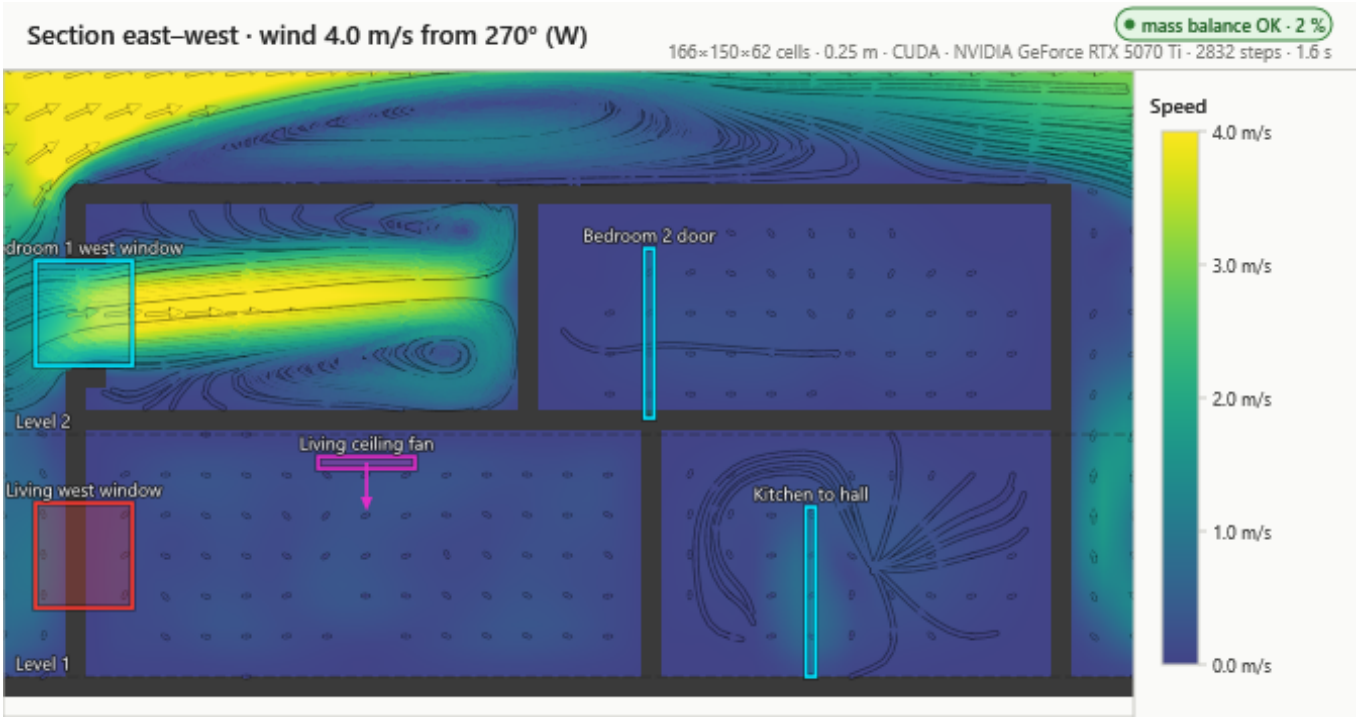


Figure 3: Section · long axis

Rooms

Level	Name	Vol m³	Voxel m³	Nominal	Fresh-air ACH	Total exch.	Thru open.	Speed m/s	Stagnant %	CO2 ppm	Age occ. min	Air-change eff.
L1	Living	156	154	yes	10.7	27.5	26.3	0.2	51 %	446	5.5	–
L1	Kitchen	61	57	yes	20.6	73.2	70.5	0.2	43 %	429	2.7	–
L1	Hall	46	46	yes	19.6	71.6	43.1	0.1	57 %	425	3.6	–
L2	Bedroom 1	113	99	yes	99.0	124.8	118.0	0.6	19 %	424	0.6	0.41
L2	Bedroom 2	74	63	yes	8.9	13.9	6.8	0.1	68 %	450	6.5	0.48
L2	Landing	30	28	yes	14.1	38.6	35.4	0.1	56 %	430	4.0	–
L2	Bathroom	25	21	yes	sealed	0.0	0.0	0.0	100 %	420	sealed	–

Fresh-air ACH = 3600 ÷ the room's mean age of air. Total exch. = all air entering the room, including from other rooms.
Definitions and limits: see How WindFlow works (how-windflow-works.pdf).

Openings

Name	Level	Kind	Status	In m³/s	Out m³/s
Living south window	L1	Window	Open	0.00	0.55
Living north window	L1	Window	Open	0.00	0.59
Living west window	L1	Window	Closed	0.00	0.00
Kitchen window	L1	Window	Open	0.34	0.00
Hall window	L1	Window	Open	0.56	0.00
Front door	L1	Door	Closed	0.00	0.00
Bedroom 1 south window	L2	Window	Open	0.00	1.71
Bedroom 1 north window	L2	Window	Open	0.00	1.78
Bedroom 1 west window	L2	Window	Open	3.71	0.00
Bedroom 2 window	L2	Window	Open	0.12	0.00
Bathroom window	L2	Window	Closed	0.00	0.00
Living to kitchen	L1	Door	Open	1.14	0.00
Kitchen to hall	L1	Door	Open	0.85	0.00
Bedroom 1 door	L2	Door	Open	0.00	0.17
Bedroom 2 door	L2	Door	Open	0.00	0.11
Bathroom door	L2	Door	Closed	0.00	0.00

Sources

Name	Kind	Set m³/h	Measured m³/h
Living ceiling fan	Fan	3240	—

— = the device is switched off, so nothing is measured.

Notes

These notes say where the result is less certain. They do not mean the result is wrong.

- Age of air has no steady state: no fresh air reaches part of the building (values capped at 24 h)
- Opening Kitchen window: effective discharge coefficient 0.36 (Q -0.337 m3/s, pressure drop -0.299 Pa) is outside 0.42-0.8 - the flow here is not simple orifice flow (air sweeping along the facade, a jet across the room, or too small a pressure difference to read); treat its flow rate with care
- Opening Bedroom 2 window: effective discharge coefficient 0.28 (Q -0.116 m3/s, pressure drop 0.032 Pa) is outside 0.42-0.8 - the flow here is not simple orifice flow (air sweeping along the facade, a jet across the room, or too small a pressure difference to read); treat its flow rate with care
- 8 openings: the grid resolves Cd 0.57-0.59 instead of 0.61, so flow through them may be 3-7 % low.
- 12 openings: discharge coefficient uncertain by about ±10-28 % at this cell size; a finer quality level reduces it.
- 5 openings are only 3.3-3.9 cells across at 255 mm cells, below the 4 cells VALIDATION.md checked, so their flow is unreliable; use a finer grid: Hall window, Living to kitchen, Kitchen to hall, Bedroom 1 door, Bedroom 2 door.
- 4 rooms are gust-driven: Living (on walls parallel to the wind), Kitchen (on one facade), Hall (on one facade), Bedroom 2 (on one facade). Their outside openings are only on those walls, so they are ventilated mainly by gusts that a steady wind does not have: Standard can read low here; run

High (gusty wind) to compare.

- Wind blockage 10.5 % (guideline 3 %). This wind domain is compact (blockage 10.5 %). In a test on the sample house, room air exchange read about 8-33 % higher than in the Full domain. For a final check use Domain: Full (Settings, Expert).
- Room Hall outline crosses open air: 0.29 m3/s crosses it through plain air (a stair void, an open-plan edge or an outline drawn through open air); it is in the total exchange, not in the 0.631 m3/s through openings